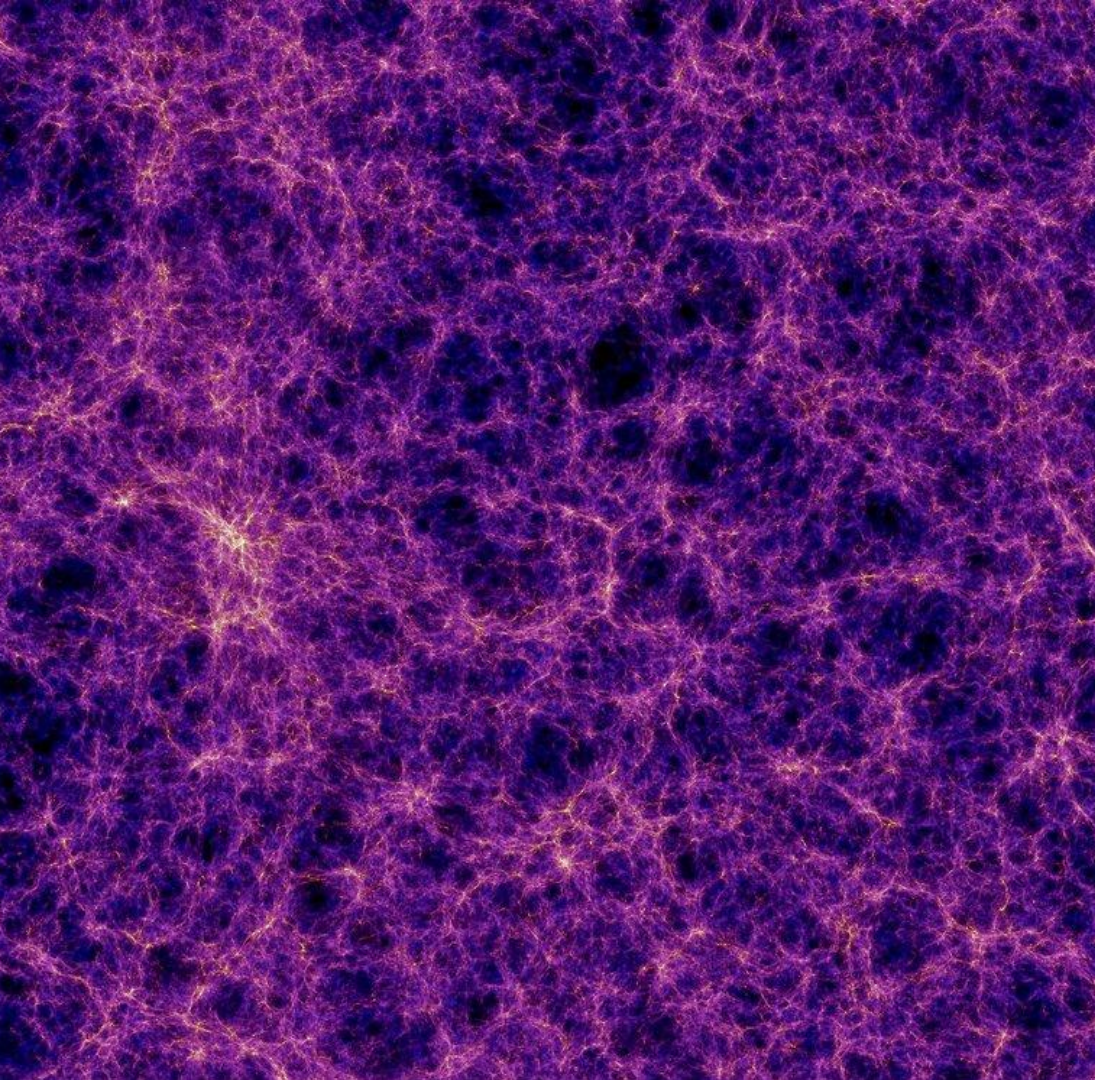




Gravitational Simulations at Cosmic Scales

Lev Kruglyak

The Particle-Mesh Algorithm

Solve the gravitational Poisson equation: $\nabla^2 \phi(x) = -4\pi G \rho(x)$

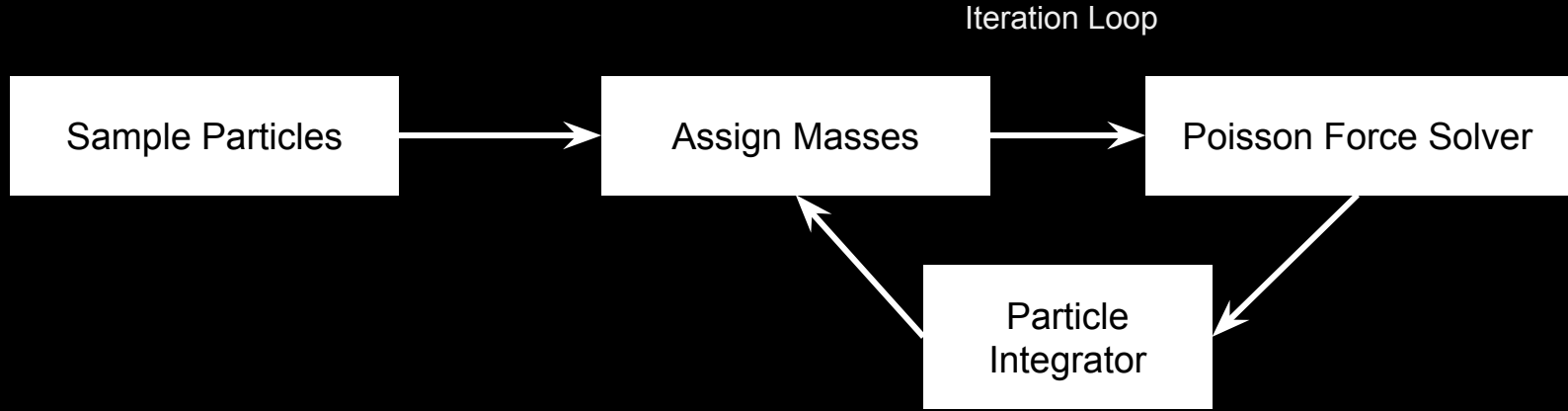
Sample particles from an initial density field

- Assign particle masses to density mesh
- Forward FT of density
- Invert the Laplacian to get spectral potential
- Compute spectral gradient of potential
- Inverse FT to get force vector field
- Integrate particles along vector field
- Repeat

$$\hat{\phi}(k) = \frac{4\pi G}{|k|^2} \hat{\rho}(k)$$

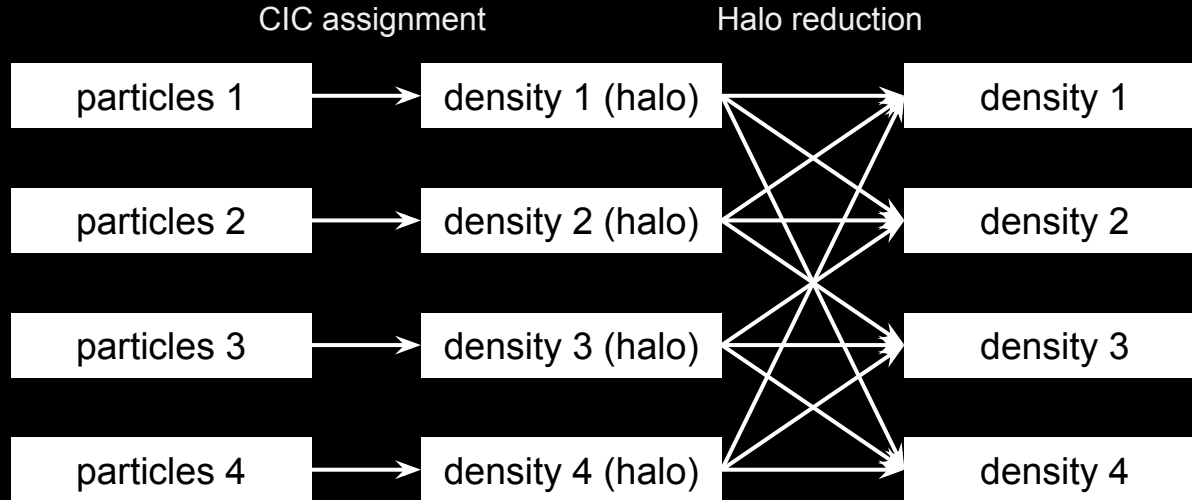
$$\widehat{\nabla_i \phi}(k) = -i k_i \partial_i \hat{\phi}(k)$$

Implementation

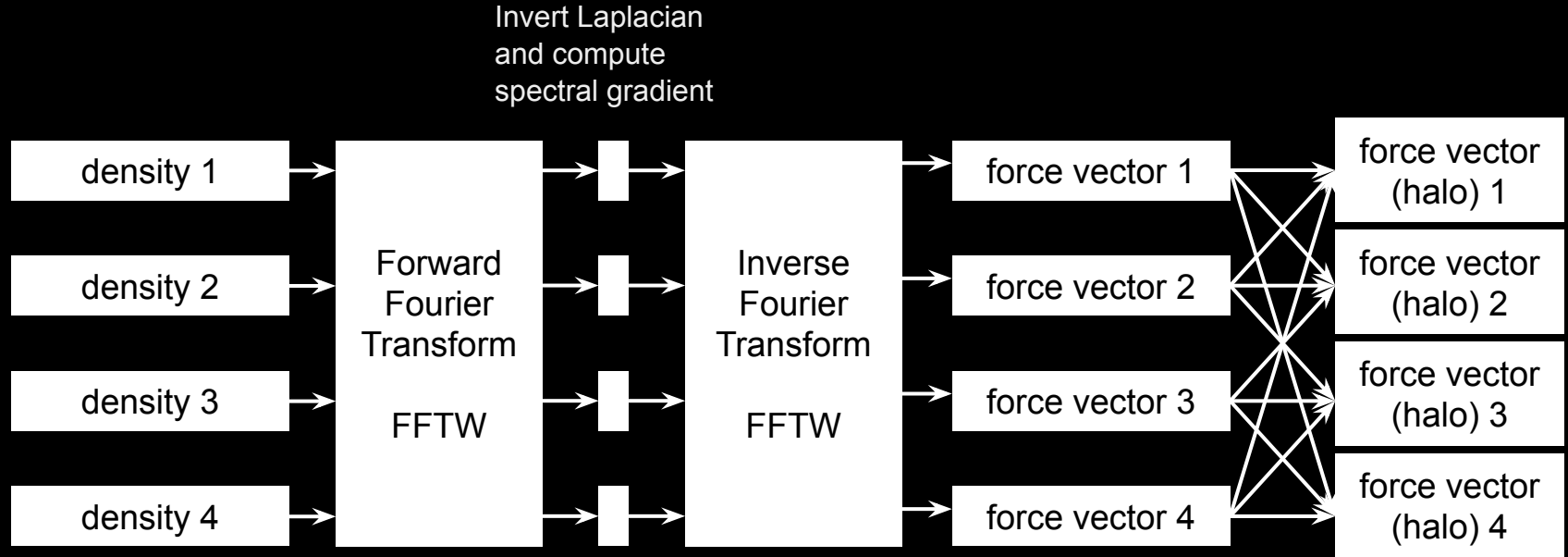


MPI: Decompose domain into horizontal strips, each managed by separate node.

Mass Assignment

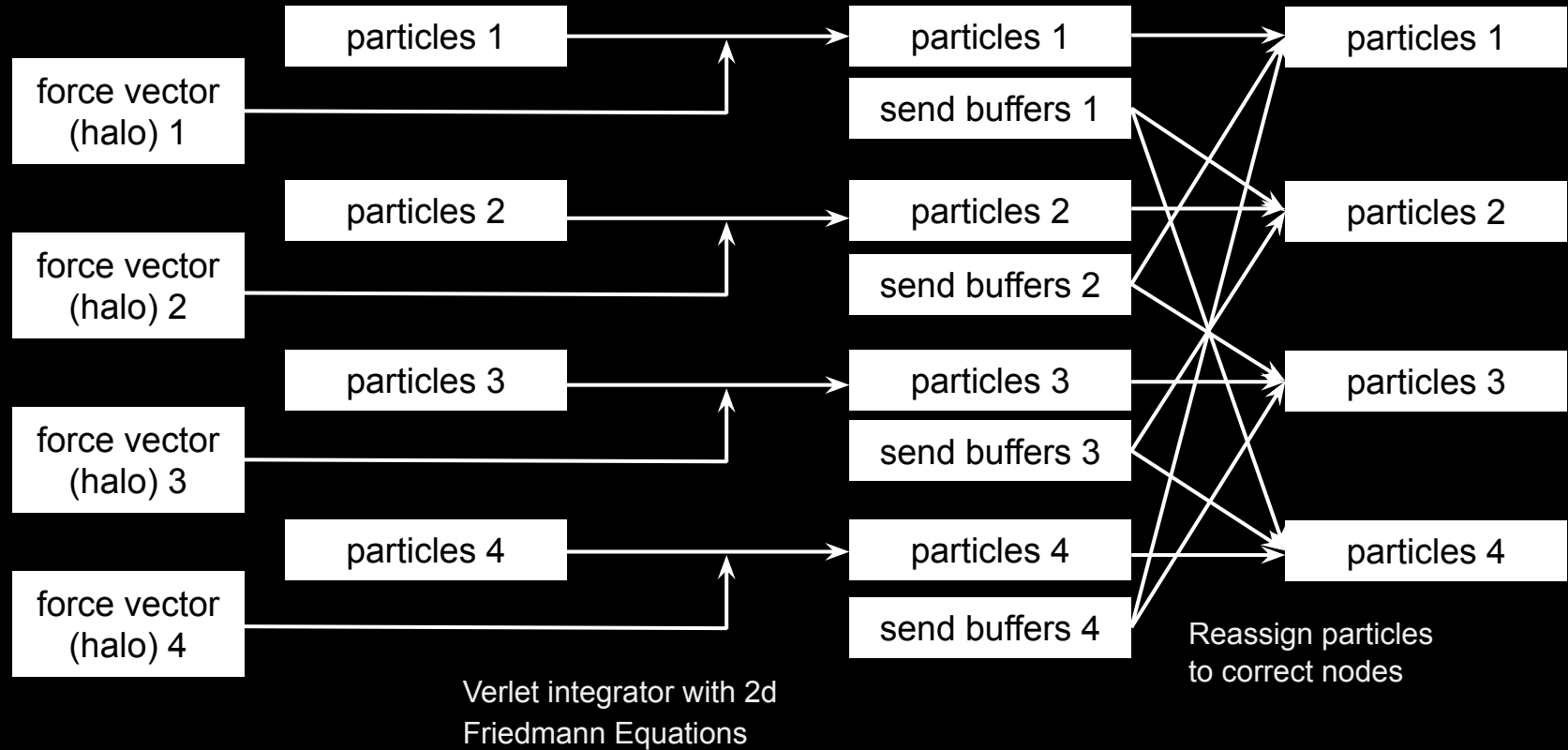


Poisson Force Solver



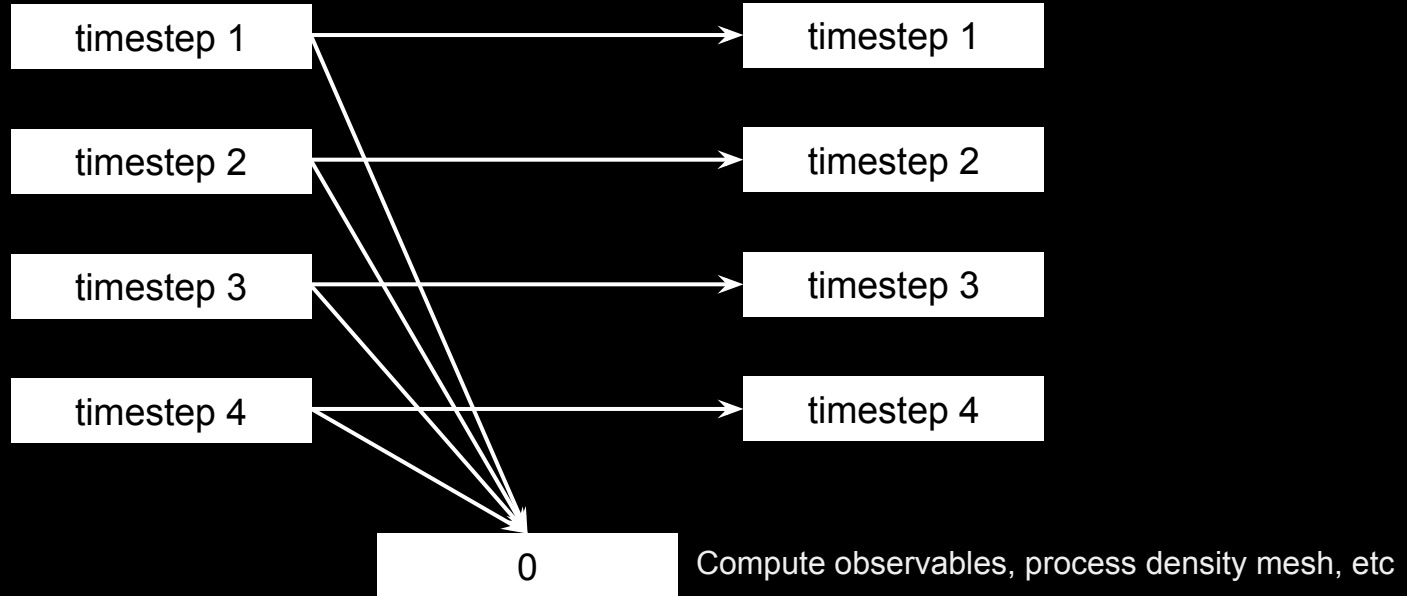
* buffer details and optimizations not shown for simplicity

Particle Integrator



* assumes low velocity

Getting Profiler Data, Sending Commands, Etc...

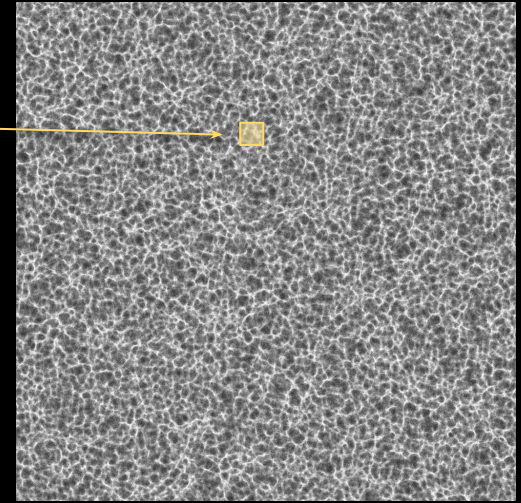


Surprise:
Live Demo!

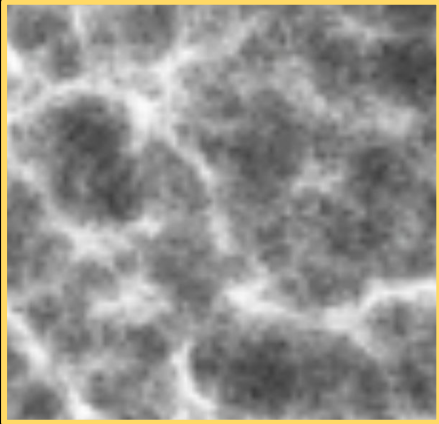
Numerical Consistency

Shows consistent results across resolutions,
number of MPI processes, number of threads,
integration parameters, etc.

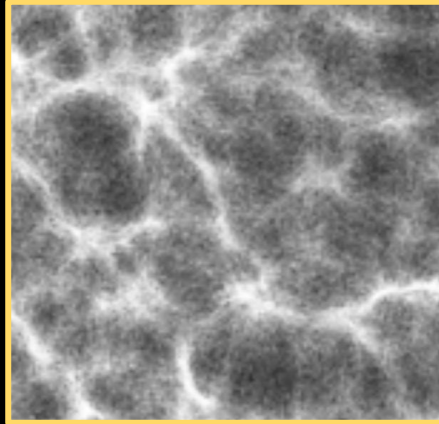
Full Density



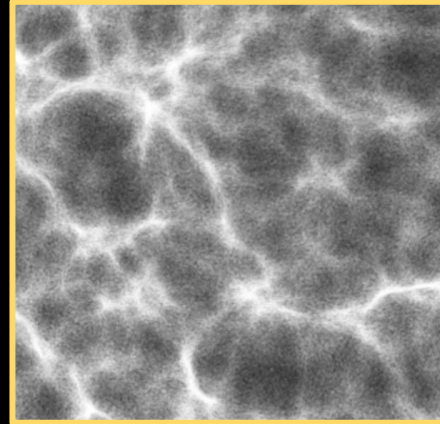
Resolution = 1024



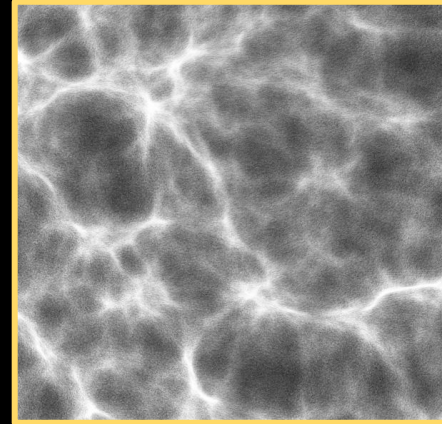
Resolution = 2048



Resolution = 4096



Resolution = 8192



3.3 Billion Particle Simulation

Density Mesh Dimension: 8192

Particles Per Mesh Cell: 50

Total Number of Particles: 3.3 billion

12 nodes, 2 tasks per node, 8 CPUs per task

Runtime per timestep: 16 seconds

Total Particle Data: 200 Gb

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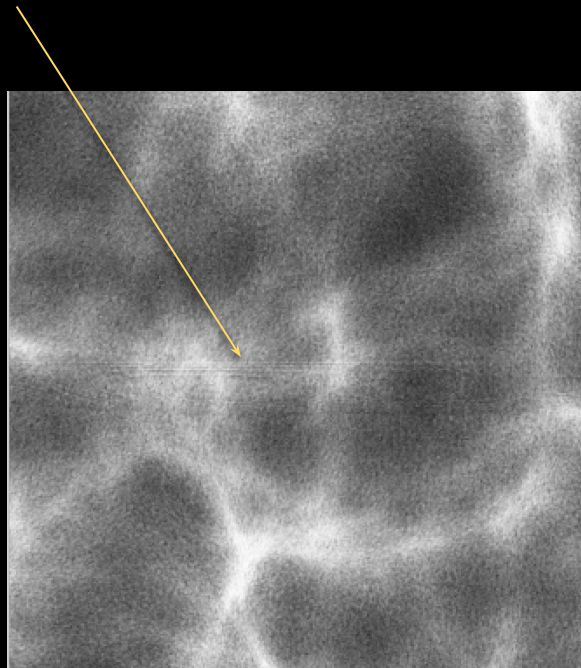
12 nodes, 2 tasks per node, 8 CPUs per task

Runtime per timestep: 16 seconds

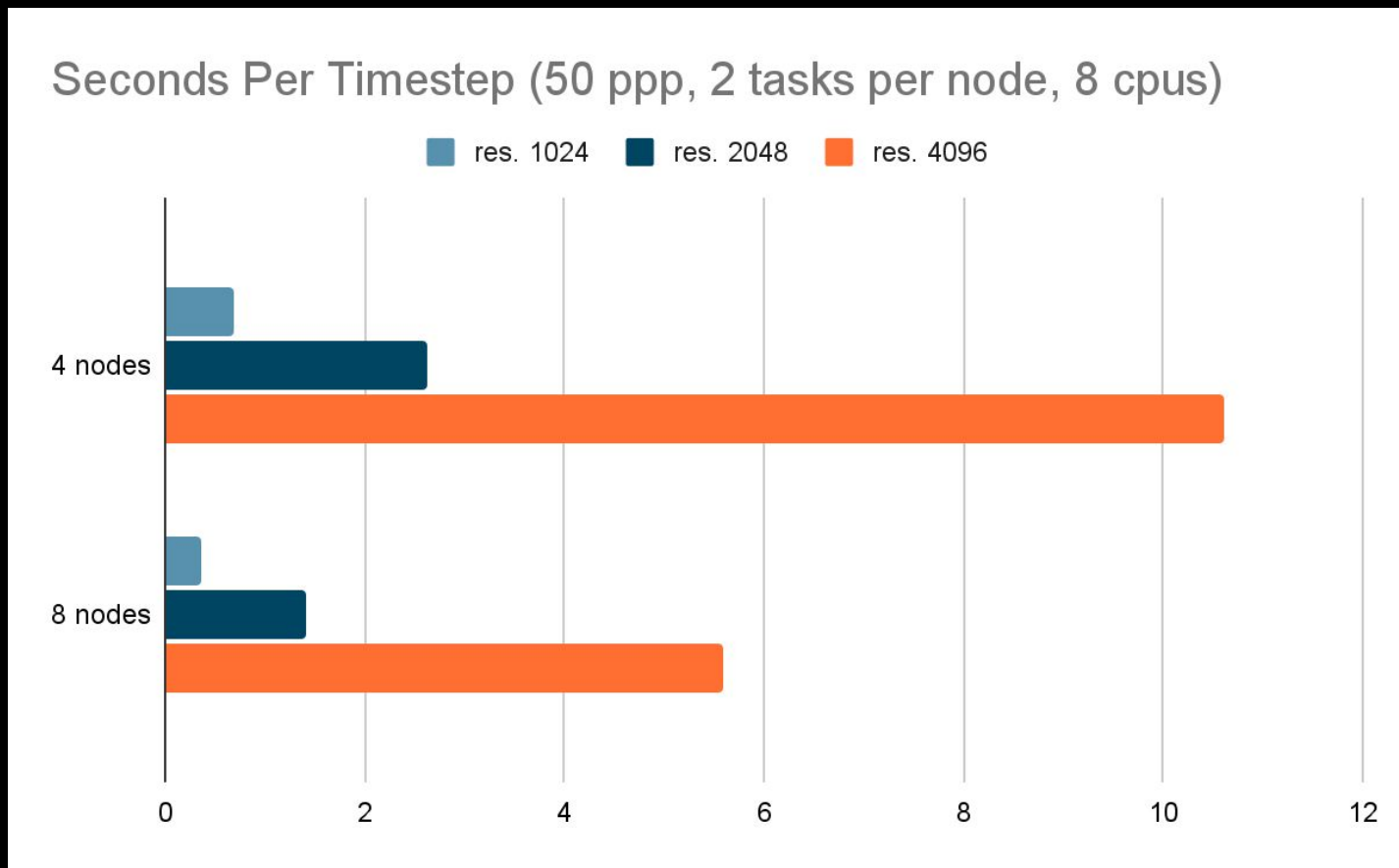
Total Particle Data: 200 Gb

Minor Issue:

Shadow of MPI boundary at over 100x zoom



Demonstration of Scaling Laws (MPI)



Further Work

- P3M algorithm - Particle-Particle Particle Mesh
- Optimizations
 - FFTW wisdom plans
 - Spatial subdivision structure for particles (better cache locality)
 - etc...